Detection, Prevalence and User Perception of Dark Patterns in E-Commerce of Bangladesh

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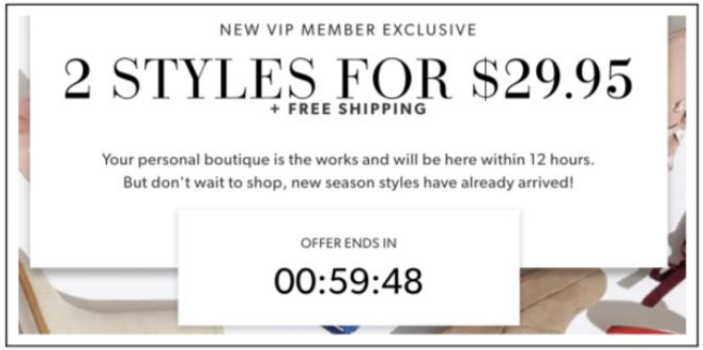
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What are ‘**Dark Patterns**’?

Dark patterns (DP) are deceptive UI design tactics that abuse knowledge of psychology to alter user decision-making [1]

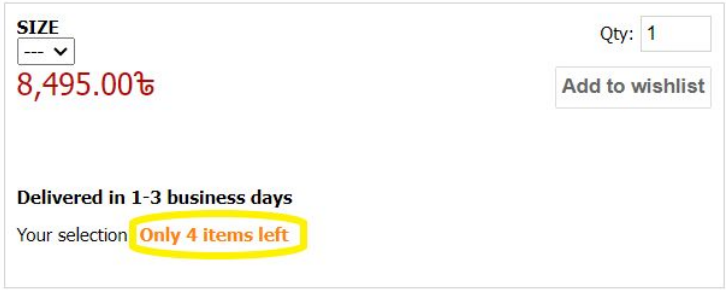
Examples of Dark Patterns



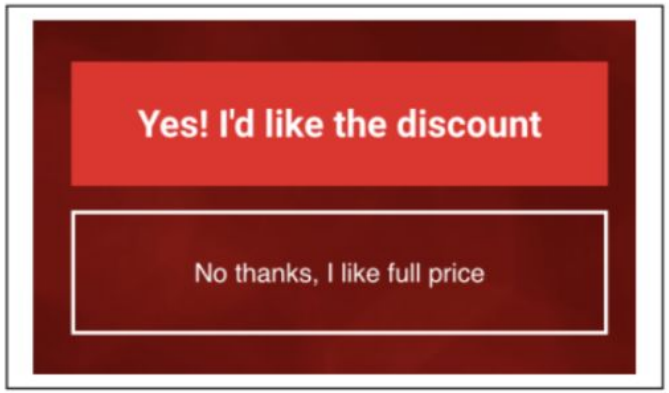
Urgency



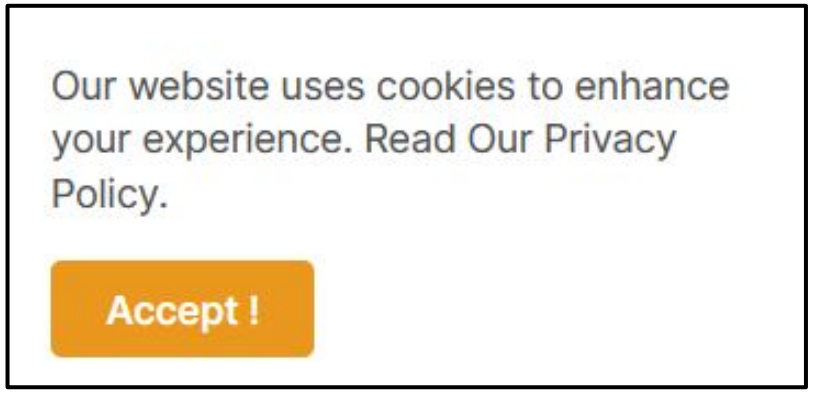
Social Proof



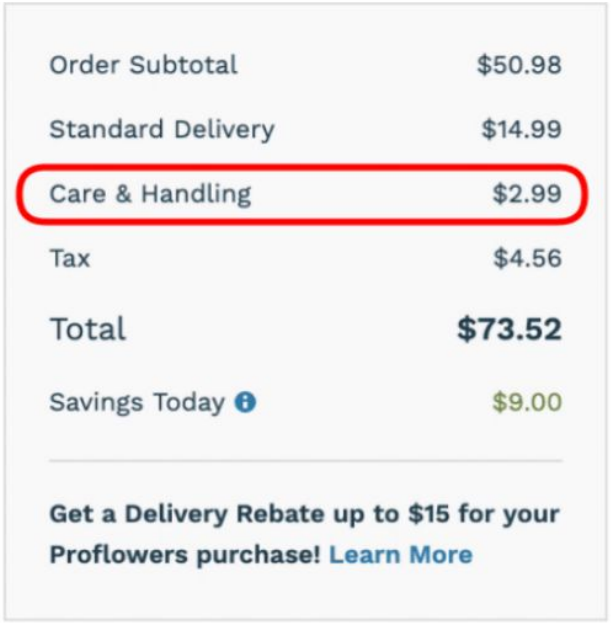
Scarcity



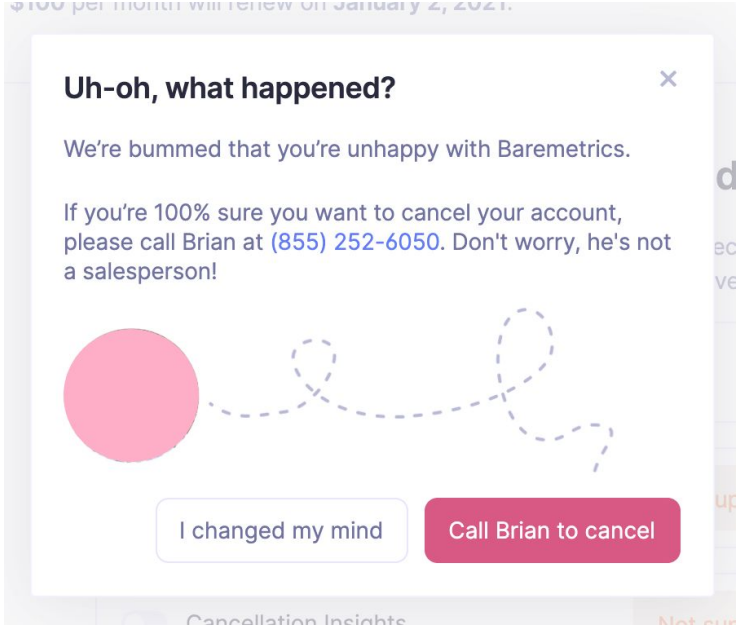
Misdirection



Forced Action



Sneaking



Obstruction

Motivation

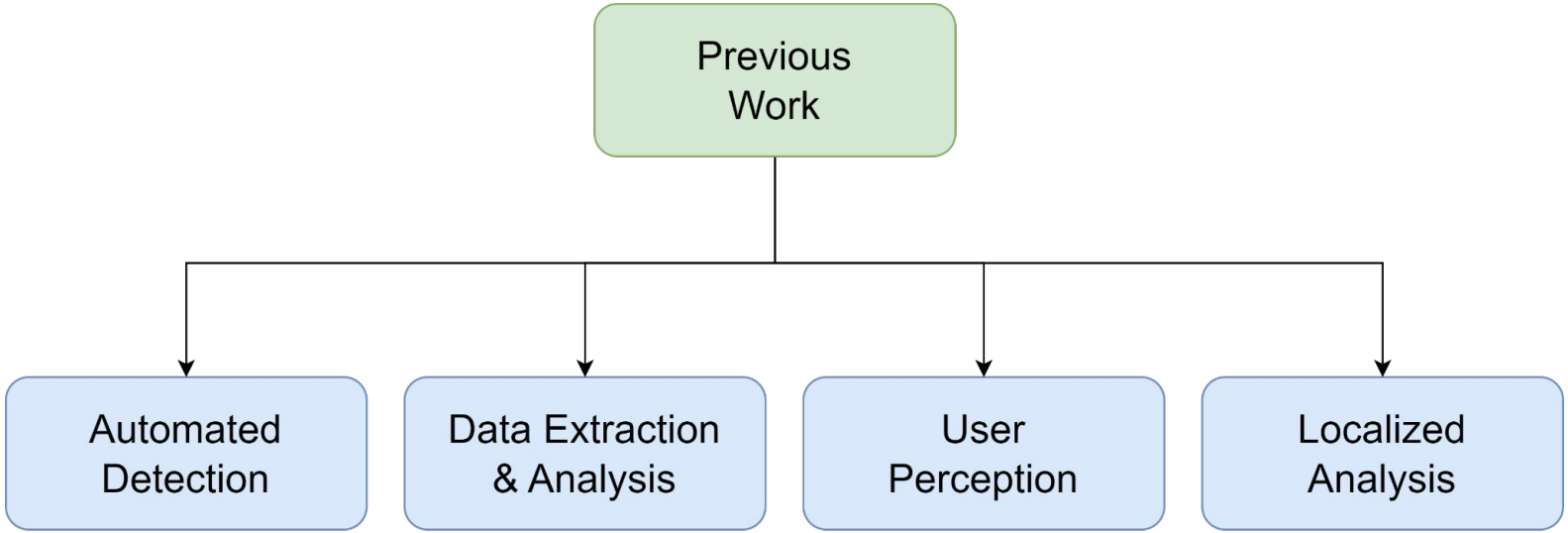
Prevalence and Impact

- Dark patterns are widespread across various digital platforms
  - **Social media** [10]
  - **E-commerce** - Mathur et al. found **dark patterns in 1,254 e-commerce websites** [2]
  - **Mobile apps** [11]
  - **Video games** [12]
- Dark patterns can manipulate users into decisions that are not in their best interests [2]
  - **Financial loss**
  - **Unnecessary sharing of personal data**
  - **Induction of addiction**

Countermeasures

- Developing **countermeasures** is important for user welfare [7]
  - Designer **guidelines**
  - User **awareness** and **education**
  - **Regulatory policy** making

Previous Work


Previous Work
Automated Detection
Data Extraction & Analysis
User Perception
Localized Analysis

Automated Detection

Authors	Detection Technique	Automation Type	Weakness
Mathur et al. [2]	Manual	Semi	Automated extraction of texts, but **manual detection**
Yada et al. [3]	Supervised ML	Full	Poor accuracy on real life data because of **data overfitting**
Soe et al. [4]	Supervised ML	Semi	**Manual extraction** of features
Mansur et al. [5]	Heuristics	Full	**Does not work** on **semantically complex**, and semantically similar yet lexically varied texts

Formats of Data Extraction



Video Recordings (Geronimo et al., 2020)



Screenshots (Mansur et al., 2023)



Extracted features (Soe et al., 2022)



Textual data (Mathur et al., 2019)

Textual Data Extraction & Analysis

- Mathur et al. collected the first large-scale textual dark pattern data [2]
  - Automated extraction of meaningful texts from webpage HTML
  - **Manual detection** of dark patterns and labelling of texts
  - **1,818 dark pattern texts** found from **11,000 shopping** websites
- Yada et al. added non dark pattern texts to this dataset [3]
  - **1,178 non dark pattern texts** were added from the same websites

User Perception

Authors	User Perception Variable	Findings
Di Geronimo et al. [6]	Awareness, Recognition	● Users often **struggle to identify** dark patterns ● **Awareness mechanisms** need to be improved
Bongard-Blanchy et al. [7]	Awareness, Concern, Recognition, Resistance	● Users recognize dark patterns, but they are often **unaware of the actual harm** ● **Demographics** can influence user perception
Gray et al. [13]	Awareness	● Users often sense that **something is "off"**, even when they can not explain it

Localized Analysis

- Hidaka et al. highlighted **differences** between **Western** and **Eastern** contexts [9]
  - Most research on dark patterns have been conducted in **Europe** or the **USA**
  - **Limited studies** on dark patterns **outside Western contexts**
  - **Localized studies** can produce **novel insights**
    - **Cultural** differences
    - **Linguistic** differences
  - **Japanese** applications showed **fewer dark patterns** compared to **American** ones

Summary of Previous Work

- Lack of a **detection** technique that is
  - **Generalized**
  - **Automated**
- **Localized analyses** can produce **novel insights**
- **No analysis** has been previously done on dark patterns in **Bangladeshi** websites
  - **Prevalence**
  - **User perception**

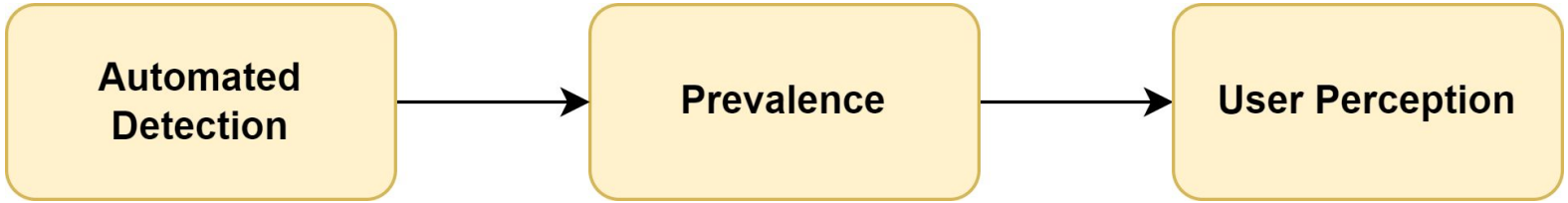
Research Questions

RQ: How can **dark patterns** be effectively **detected**, **classified**, and **analyzed** in terms of their **prevalence** and **user perception** in **Bangladeshi e-commerce** websites?

SQ1: How can dark patterns be **effectively detected** and **classified** in the context of Bangladesh?

SQ2: How **prevalent** are dark patterns in **Bangladeshi e-commerce** websites, and how do **users perceive** them?

Overview of Methodology



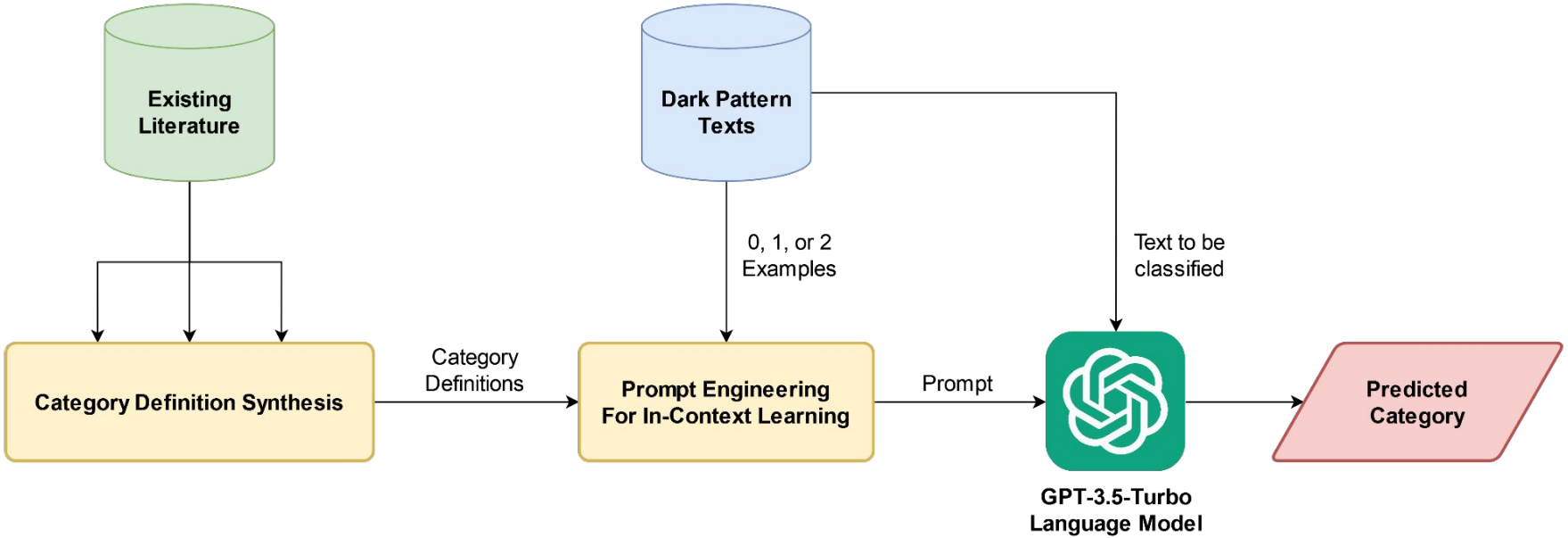
Methodology

Automated Detection of Dark Patterns

In-Context Learning

- An **emergent** ability of large language models (LLMs) [14]
- LLMs can pick up language semantics **without requiring any training**
    - Example: GPT-3 Can label a text as ‘racist’, ‘sexist’, or ‘hate speech’ [15]
- **Context** and **Examples** can be provided **within the prompt**
- Underlying hyperparameters of the model do not change [16]
- **No training → No dependency on data → Increased generalizability**

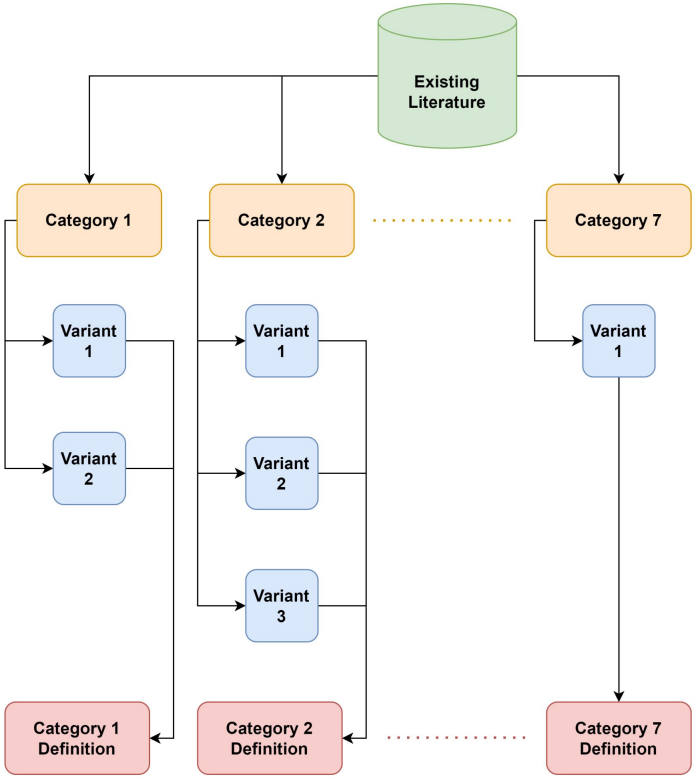
Proposed Approach for Automated Detection



Category Definition Synthesis

- **7 broad categories** of dark patterns [2]
  - Misdirection, Urgency, Social Proof, Forced Action, Obstruction, Sneaking, and Scarcity
- Each category may have **multiple variants**
  - Urgency - Countdown Timer, and Limited-time Message
  - Social Proof - Activity Message, and Testimonial
  - Scarcity - Low-stock Message, and High-demand Message
  - and so on

Category Definition Synthesis



Output of Category Definition Synthesis

Classification Category	Definition
Misdirection	‘Misdirection’ includes ‘attracting the user’s attention to one thing in order to distract their attention from another’ and ‘use of visuals, confusing language, and emotion (shame) to intrigue users to or away from particular choices’.
Urgency	‘Urgency’ means ‘pressuring the user to accept deals or discounts with time limitations, count-down timers, or sale deadlines’.
Scarcity	‘Scarcity’ means ‘pressuring the user to accept deals or discounts with indications of low stock, limited supply, high demand, or low availability’.
Social Proof	‘Social Proof’ includes ‘informing the user about the activity on the website (e.g., purchases, views, visits) to make the product more credible’ and ‘testimonials on a product page whose origin is unclear’.

Output of Category Definition Synthesis

Classification Category	Definition
Obstruction	‘Obstruction’ includes ‘making a process more difficult than it needs to be, with the intent of obstructing certain actions’ and ‘making it easy for the user to sign up or subscribe for a service but hard to cancel it’.
Forced Action	‘Forced Action’ means ‘requiring the user to perform an undesirable action (e.g., creating accounts or sharing information) to access certain functionality’.
Sneaking	‘Sneaking’ includes ‘hiding or delaying the relevant information from the user’, ‘adding additional products to shopping carts without their consent’, ‘revealing previously undisclosed charges to users right before the checkout’ and ‘enrolling users in a recurring subscription or payment plan without clear disclosure or their explicit consent’.
Not Dark Pattern	If a text does not fall into the other 7 categories, it is defined as ‘Not Dark Pattern’.

Prompt Engineering for In-Context Learning

- Prompting techniques used in this study
  - Zero shot prompting
    - **No example** is used
  - One shot prompting
    - **Only one example** per category is used
  - Few shot prompting
    - **Two examples** per category are used

Zero Shot Prompting

Prompt:

Context <Definition of Dark Patterns> <Classification of Dark Patterns>

<Definition of Category 1> ~~<Example 1 of Category 1> <Example 2 of Category 1>~~

.....

<Definition of Category 7> ~~<Example 1 of Category 7> <Example 2 of Category 7>~~

<Definition of Category 8 (Not Dark Pattern)>

Input <Input Format> <Processing Instruction>

Output <Output Format> <Text to be classified>

GPT-3 Response: <Predicted Category>

One Shot Prompting

Prompt:

Context <Definition of Dark Patterns> <Classification of Dark Patterns>

<Definition of Category 1> <Example 1 of Category 1> ~~<Example 2 of Category 1>~~

.....

<Definition of Category 7> <Example 1 of Category 7> ~~<Example 2 of Category 7>~~

<Definition of Category 8 (Not Dark Pattern)>

Input <Input Format> <Processing Instruction>

Output <Output Format> <Text to be classified>

GPT-3 Response: <Predicted Category>

Few Shot Prompting

Prompt:

Context <Definition of Dark Patterns> <Classification of Dark Patterns>

<Definition of Category 1> <Example 1 of Category 1> <Example 2 of Category 1>

.....

<Definition of Category 7> <Example 1 of Category 7> <Example 2 of Category 7>

<Definition of Category 8 (Not Dark Pattern)>

Input <Input Format> <Processing Instruction>

Output <Output Format> <Text to be classified>

GPT-3 Response: <Predicted Category>

Examples of Dark Pattern Texts

Dark Pattern Category	Examples
Misdirection	1. No, I'll rather pay full price 2. No thanks, I dont want a discount
Urgency	1. Items reserved for 15:00 2. 1 day 08:15:25
Scarcity	1. 1 LEFT 2. 87% offers claimed. Hurry up!
Social Proof	1. IN 43 PEOPLE'S SHOPPING BAG 2. Armin Dinovic bought 10M Runescape 3 Gold Order total: 3,70€ About 10 seconds ago

Experiments

Automated Detection of Dark Patterns

Validation Dataset

- Combined dark pattern text dataset of Mathur et al. [2] and Yada et al. [3]
  - Data collected from **11,000 e-commerce websites**
  - **2,356 instances**
    - **1,178 (50%) dark pattern** instances
      - Labelled with **7 broad categories** of dark patterns
    - **1,178 (50%) non-dark pattern** instances
- **Examples used** in prompts were **removed** from **validation dataset**

Experimental Results

Category	Accuracy		
	Zero Shot	One Shot	Few Shot
Misdirection	6.67%	53.61%	65.8%
Urgency	93.33%	92.34%	93.27%
Scarcity	92.34%	95.92%	98.56%
Social Proof	92.63%	96.46%	94.84%
Obstruction	11.11%	84.62%	92%
Forced Action	100%	100%	100%
Sneaking	0%	0%	0%
Not Dark Pattern	91.51%	94.65%	94.91%
Overall	83.57%	90.97%	92.57%

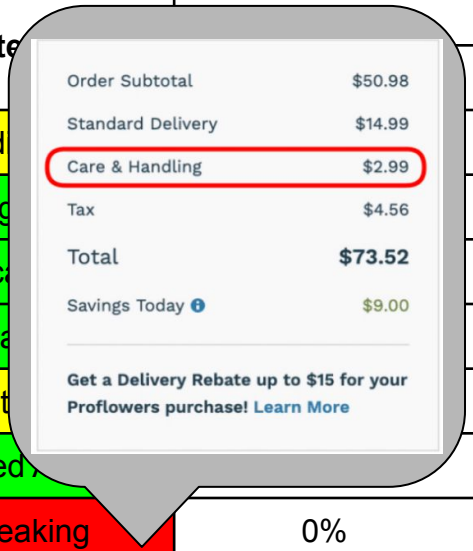
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Obstr		84.62%	92%
Forced		100%	100%
Sneaking	0%	0%	0%
Not Dark Pattern	91.51%	94.65%	94.91%
Overall	83.57%	90.97%	92.57%

Textual analysis is not enough

Experimental Results

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Forced		100%	100%
Sneaking	0%	0%	0%
Not Dark Pattern	91.51%	94.65%	94.91%
Overall	83.57%	90.97%	92.57%



Experimental Results with Weaker Definitions

	Zero-Shot	**Few-Shot**
Accuracy Using **Synthesized Definitions**	83.57%	92.57%
Accuracy Using **Weaker Definitions**	82%	90.35%
P-Value (McNemar's Test)	0.0094	4.9228e-06
H_0 Rejected?	Yes	Yes

Experimental Results with Weaker Definitions

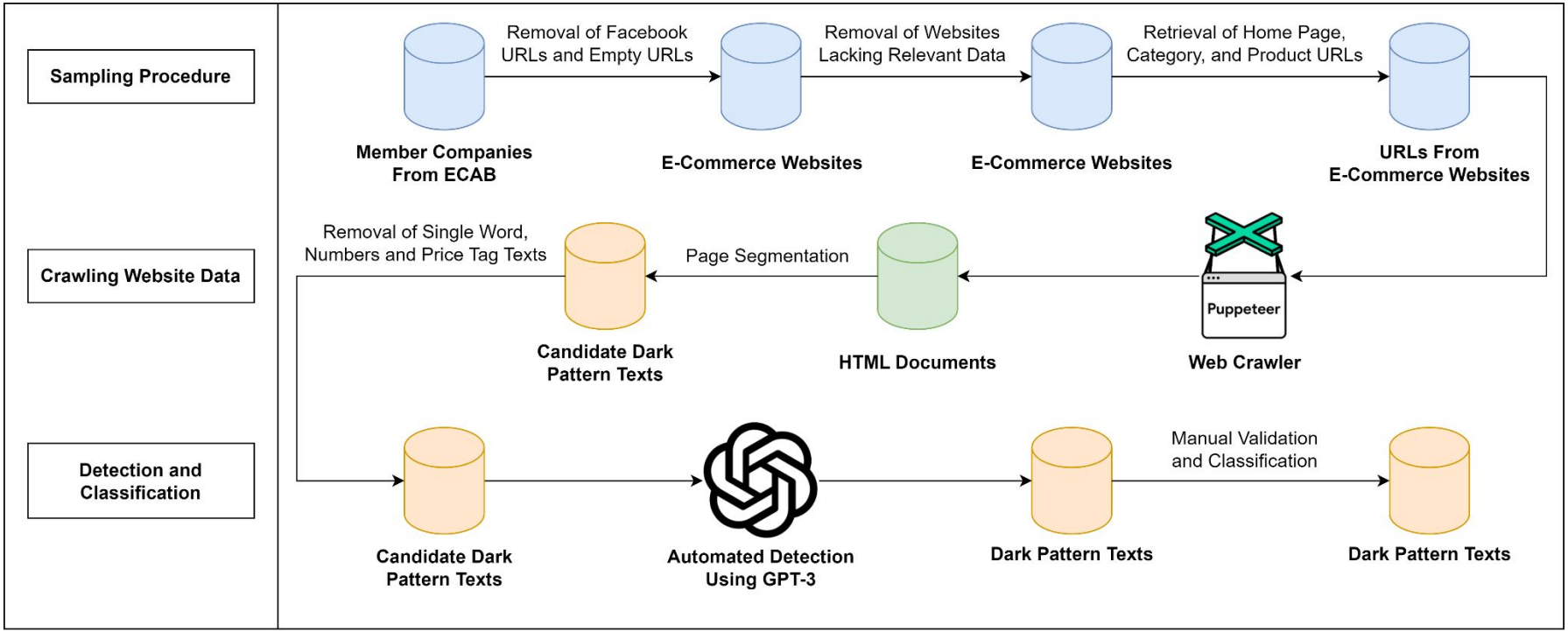
	Zero-Shot	**Few-Shot**
Accuracy Using **Synthesized Definitions**	83.57%	92.57%
Accuracy Using **Weaker Definitions**	82%	90.35%
P-Value (McNemar's Test)	0.0094	4.9228e-06
H_0 Rejected?	Yes	Yes

So, better definitions can generate better performance

Methodology

Prevalence of Dark Patterns

Extraction of Dark Patterns to Reveal Prevalence





- 2,216 member companies
- Largest publicly available set of e-commerce companies
- Best possible sampling frame for e-commerce websites
- Contains member profiles
  - Company Name
  - Website URL

Website Exclusion Criteria

- Facebook page URLs
- Empty URLs
- Webpages with no further navigation
- Websites lacking relevant data
- Websites with connection problems
- Final sample consists of **2,068 webpages** from **715 websites**

Crawling Website Data

- JavaScript library '**puppeteer**' is used
- Whole HTML documents are retrieved
- Some websites block web crawlers
  - JavaScript library '**puppeteer-extra-plugin-stealth**' is used
  - Conceals properties from outgoing requests that can flag the requests as crawlers

Page Segmentation Algorithm

- Text segments need to be extracted from HTML documents
- Initially used **segmentation algorithm** proposed by **Mathur et al.** [2]
  - Resulted in the discovery of **very few dark patterns**
  - Algorithm was too restrictive, **discarding potential dark pattern texts**
- We **relaxed** the segmentation algorithm
  - **Any visible HTML element that is a leaf and contains text content**
  - **Paragraph elements were excluded** as they usually represent longer blocks of descriptive text
  - Resulted in the discovery of an **increased number of dark patterns**
- 770,570 text segments were extracted

Algorithm 1: Page Segmentation.

```
   Input: HTML Document
   Output: List of Text Segments
1  Function extractTextSegments(page):
2      ignoredElements ← ["p", "script",
         "style", "noscript", "br", "hr"];
3      segmentsList ← [];
4      foreach element in page do
5          if element is not in ignoredElements
             and element is visible and element
             does not contain any child elements
             then
6              innerText ← element.innerText;
7              segmentsList.append(innerText);
8          end
9      end
10     return segmentsList;
```

Proposed Page Segmentation Algorithm

Automated Detection

- Data Preprocessing
  - **Removed** texts that consisted of only **numbers** or a **single word**
  - **Removed price tag** texts containing the local currency sign, using **regular expressions**
- Initially used best-performing model (**RoBERTalarge**) of **Yada et al.** [3]
  - Resulted in a **large number of false positives**
  - Manual verification would be lengthy, undermining the purpose of automated detection
  - Model was vulnerable to **data overfitting**
  - Few-shot model of our **proposed approach** resulted in around **90% less false positives**
- **Manual classification** was carried out to discard the remaining false positives

Our Approach vs RoBERTalarge of Yada et al. [3]

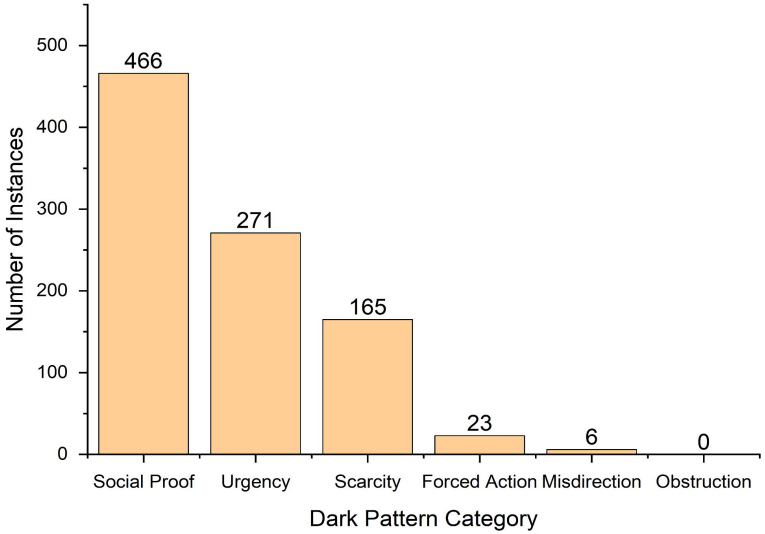
- 198 text instances were manually labelled to compare the two models
  - 99 (50%) dark pattern texts
  - 99 (50%) non-dark pattern texts (randomly selected)
- **McNemar's test** was carried out for significance testing
- P value → **2.615e-12** → H_0 rejected → **Significant difference**
- Accuracy
  - **RoBERTalarge: 68%**
  - **Our approach: 98%**

Findings

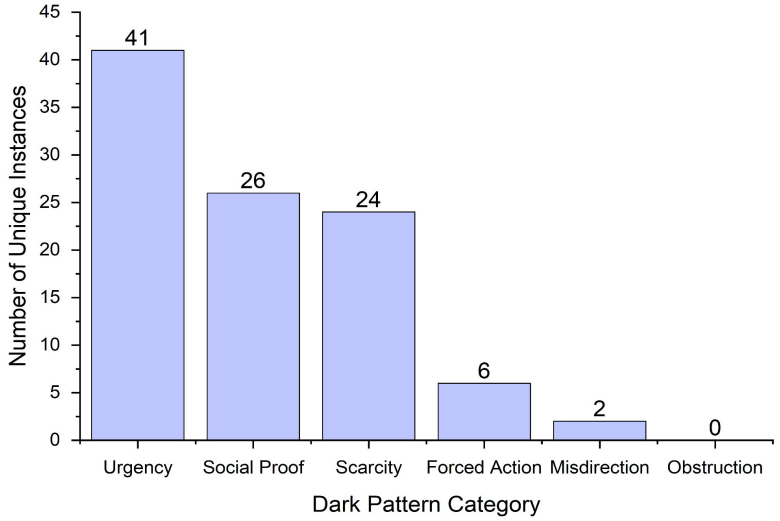
Prevalence of Dark Patterns

Prevalence of Dark Patterns in Bangladesh

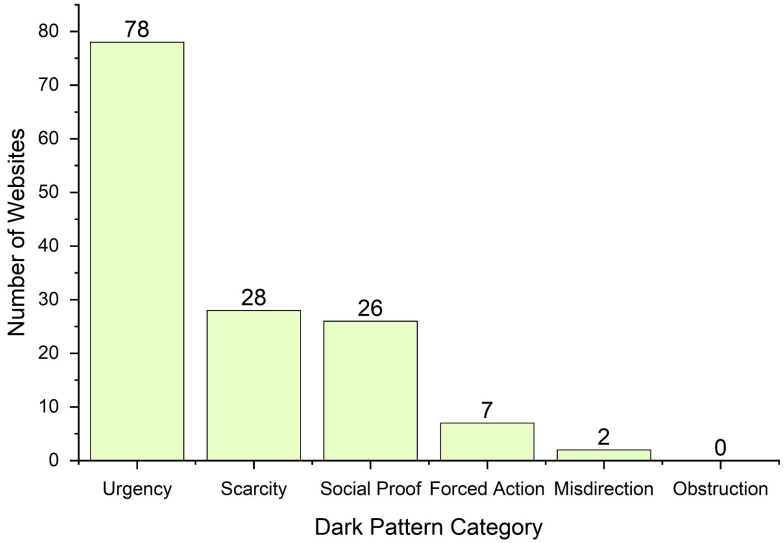
- Dark patterns were found in **131 (18.3%) out of 715 websites**
    - **Higher rate** than what **Mathur et al.** found globally (**11.1% of 11,000 websites**) [2]
- **931 dark pattern** instances were found
- **99 unique** instances
- **Social Proof**, **Urgency**, and **Scarcity** were the **most prominent** categories
- Forced Action, Misdirection, and Obstruction categories were rarely found
- No correlation in presence of dark patterns from different categories



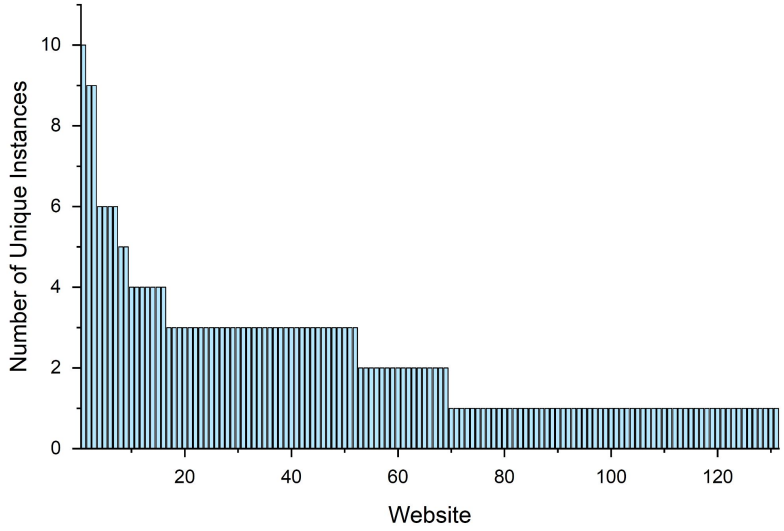
Number of Instances Across DP Categories



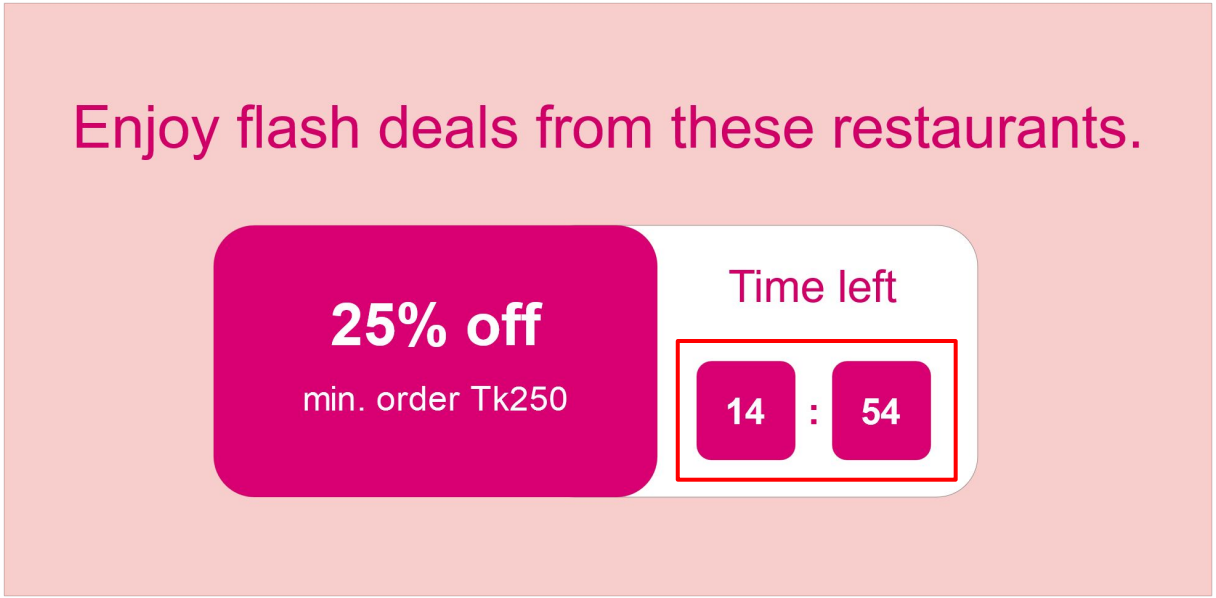
Number of Unique Instances Across DP Categories



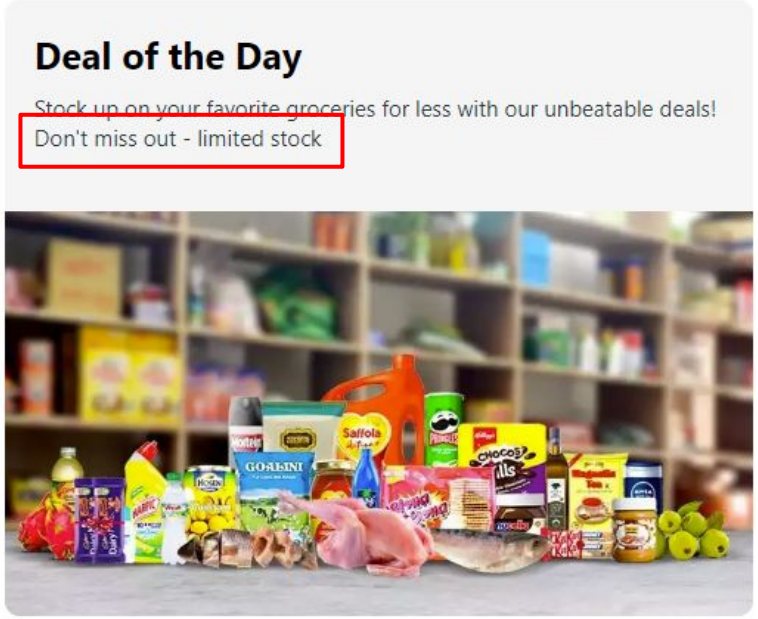
Number of Websites Across DP Categories



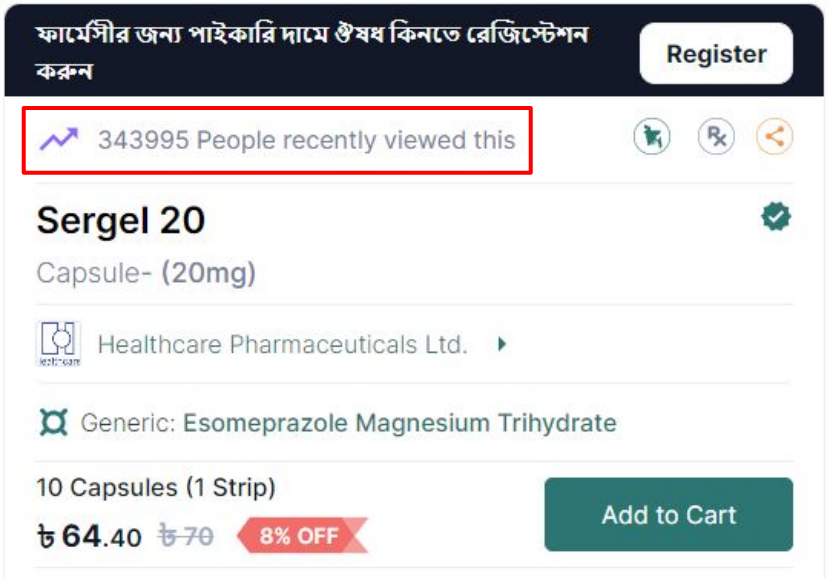
Number of Unique Instances Across Websites



Example of ‘Urgency’ Dark Pattern in Foodpanda



Example of ‘Scarcity’ Dark Pattern in Chaldal



Example of ‘Social Proof’ Dark Pattern in Arogga

Methodology

User Perception of Dark Patterns

Study Design - Goal Question Metric (GQM)

- **Goal**: Analyze **user perception** of dark patterns in **Bangladeshi e-commerce** websites
- **Questions**
  - How many users have **encountered** each type of dark pattern?
  - To what extent are users **aware** that these designs can **influence their actions**?
  - How **concerned** are users about the **influence** of these designs on **decision-making**?
- **Metrics**: Average ratings of **Exposure**, **Awareness**, and **Concern**

Study Design - Survey Content

- UI design instances
  - 6 instances of **dark patterns**
  - 4 instances of **non-dark patterns** (to counteract negative perception)
  - **Randomly ordered**
- **Three ratings** were measured for each design
  - **Exposure** (0 = never encountered, 1 = encountered)
  - **Awareness** (from 1 = strongly disagree to 5 = strongly agree)
  - **Concern** (from 1 = strongly disagree to 5 = strongly agree)
- Investigation of **differences in perception** based on **demographics**
  - Gender
  - **Educational background**

Study Design - Participants

- Number of participants: 68
- Demographics
  - Gender: 45 male, 23 female
  - Age range: 19 to 27 years (Mean: 22.57, SD: 1.47)
- Groups
  - Educational Background in **Computer Science** or **Software Engineering**: 36 participants
  - No Educational Background in Technology: 32 participants

Findings

User Perception of Dark Patterns

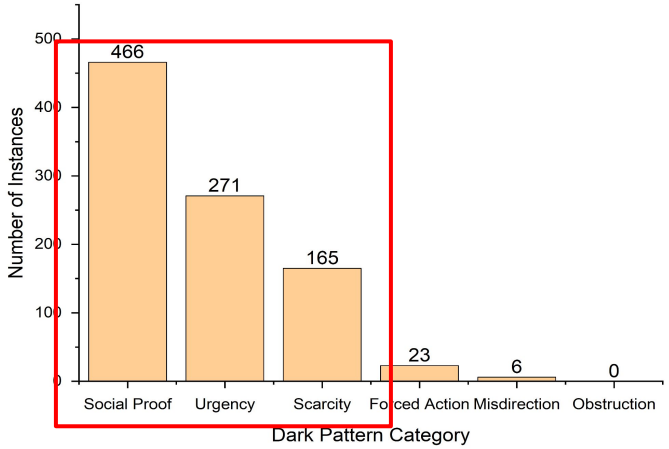
Ranking Dark Pattern Categories

Rank	Exposure		Awareness		Concern	
	DP Category	Mean Response (%)	DP Category	Mean Response (1 to 5)	DP Category	Mean Response (1 to 5)
1	Forced Action	94.12	Urgency	4.21	Obstruction	3.81
2	Urgency	88.24	Scarcity	3.99	Forced Action	3.78
3	Scarcity	86.76	Social Proof	3.96	Urgency	3.56
4	Social Proof	77.94	Obstruction	3.96	Scarcity	3.44
5	Misdirection	73.53	Forced Action	3.81	Social Proof	3.22
6	Obstruction	30.88	Misdirection	3.65	Misdirection	3.17

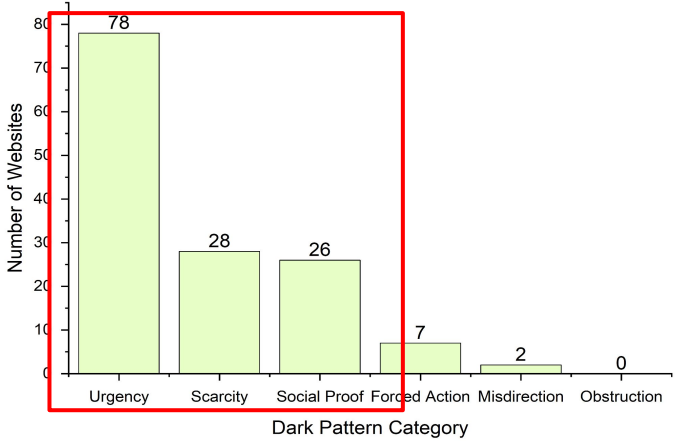
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	DP Category	Mean Response (%)	DP Category	Mean Response (1 to 5)	DP Category	Mean Response (1 to 5)
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2	Urgency	88.24	Scarcity	3.99	Forced Action	3.78
3	Scarcity	86.76	Social Proof	3.96	Urgency	3.56
4	Social Proof	77.94	Obstruction	3.96	Scarcity	3.44
5	Misdirection	73.53	Forced Action	3.81	Social Proof	3.22
6	Obstruction	30.88	Misdirection	3.65	Misdirection	3.17

Top Level Categorization of Dark Patterns



Number of Instances
Across DP Categories



Number of Websites
Across DP Categories

Top Level Categorization of Dark Patterns

Top Level Category	Member Categories	Way of Abusing User Psychology	Prevalence	User Awareness	User Concern	User Acceptance
Passive Dark Patterns	Urgency Scarcity Social Proof	Entice and Attract	Higher	Higher	Lower	Neutral
Active Dark Patterns	Obstruction Misdirection Forced Action	Mislead or Force	Lower	Lower	Higher	Negative

Perception of Tech vs Non Tech People

- Mann-Whitney U Test was used with the following hypotheses -
  - H_0: Ratings from CS participants are lower than or equal to that of NCS participants
  - H_a: Ratings from CS participants are greater than that of NCS participants

- Participants with an **educational background** in **computer science** or **software engineering** are more '**aware**' and '**concerned**' about dark patterns than participants without such a background.

Variable	Mann-Whitney U Statistic	P Value	H_0 Rejected
Exposure	614	0.3184	No
Awareness	790	0.0042	Yes
Concern	739	0.0225	Yes

Threats To Validity - Automated Detection

- Threats to External Validity
  - **Evaluation on only two datasets** poses a threat as results may vary based on test data sources
  - **GPT-3 as the LLM of choice** may also affect validity as other LLMs may generate different results
- Threats to Internal Validity
  - **Random selection of example texts** may affect internal validity
  - **Order of dark pattern categories** used in prompt engineering poses a threat to internal validity

Threats To Validity - Prevalence & User Perception

- Threats to External Validity
  - Sampling frame poses a threat as **not all e-commerce websites of Bangladesh are members of e-CAB**
  - **Analyzing a few URLs per website** also poses a threat, as other URLs may produce different results
  - **Survey participants are students**, and findings may differ for other age groups and educational levels
- Threats to Internal Validity
  - Survey **participants** can make **assumptions** about which responses are **socially desirable**
  - **Random ordering** of the **user interface design instances** also poses a threat to the internal validity
- Threats to Construct Validity
  - **Automated detection** may have missed some dark pattern data as **false negatives**
  - Survey **participants** can **misunderstand** questions or appropriate **contexts**

Conclusion

- Novel approach for **automated detection** of dark patterns
  - **In-context learning** makes the approach **generalizable**
  - Large language models
  - **Definition** of categories as **contextual** information
- **Prevalence** of dark patterns in **e-commerce** websites of **Bangladesh**
  - 18.3% e-commerce websites have at least one dark pattern instance
  - Regulatory actions needed to safeguard user welfare
- Novel **page segmentation** algorithm
  - Simpler inclusion criteria resulted in better performance

Conclusion

- Dark pattern **dataset**
  - **99** unique textual dark pattern instances
  - Labelled with **6** dark pattern categories
- **Ranking** of dark pattern categories
  - According to user **exposure**, **awareness**, and **concern**
- Novel **top level categorization** of dark patterns
  - **Active** and **Passive** dark patterns
  - According to approach of **abusing user psychology**
- **Influence of technology education** on user perception
  - Students from **computer science** and related subjects are more **aware** and **concerned**

Future Work

- **Developer Perception** of Dark Patterns
- **Legal and Regulatory** Frameworks for Bangladesh
- **Automated Detection Tool** for Dark Patterns
- **Cultural and Linguistic Differences** in Dark Patterns
- **Novel Dark Pattern Types** in Bangladesh

Publications

- Automated Detection
  - “Automated Detection of Dark Patterns Using In-Context Learning Capabilities of GPT-3” in Proceedings of the 30th Asia-Pacific Software Engineering Conference (**APSEC**), 2023, pp. 569-573.
- Prevalence and User Perception
  - “Prevalence and User Perception of Dark Patterns: A Case Study on E-Commerce Websites of Bangladesh” in Proceedings of the 19th International Conference on Evaluation of Novel Approaches to Software Engineering (**ENASE**), 2024, pp. 238-249.



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Thank You